

Composition Pair B1.01 + B1.07 — The Enabling Pair Where the Expression Mechanism Provides the Cross-Level Governed Pathway for the Reasoning Layer to Govern the Instinct Layer Across Structural Levels, Making B1.07 the Cross-Level Implementation of B1.01’s Instinct/Reasoning Separation

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its contribution is to identify and formalize the enabling relationship between the instinct/reasoning separation (B1.01) and the expression mechanism (B1.07), establishing that the two architectural commitments are not independently sufficient in a multi-level deployment and that the expression mechanism’s design is intelligible as an architectural commitment only in light of the governance requirement B1.01 imposes.

Abstract

In a single-cell deployment, the instinct/reasoning separation operates at cell scope: cell DNA (the reasoning layer) governs cell LLM behavior (the instinct layer). The expression mechanism — harness substrate activation specifying which DNA elements are active for a given goal — appears in that context as a cell-internal configuration tool whose full significance is not yet visible. In a multi-level deployment with cells composing into aspects and Selves, the situation changes. Self-level reasoning governance must reach cell-level instinct behavior through a governed pathway, not through direct cell DNA modification that bypasses that pathway. This note formalizes the enabling relationship between B1.01 (instinct/reasoning separation as independently-evolving layers) and B1.07 (expression mechanism — harness substrate activation determining which DNA elements are active per cell goal): the expression mechanism is the cross-level implementation of the instinct/reasoning separation, providing the governed channel through which Self-scope reasoning governance propagates to cell-scope instinct behavior. B1.07 enables B1.01’s cross-level operation; B1.01 motivates why B1.07 must be a governed pathway rather than a technical convenience. Without their composition, multi-level deployments degrade either to cell-scope-isolated governance (the governed cross-level channel is absent) or to expression without governance rationale (the channel exists but is misconfigured as a technical feature rather than as a governed extension of the reasoning layer).

1. Pair identification

Pair: B1.01 (the instinct/reasoning separation as independently-evolving layers) and B1.07 (the expression mechanism — harness substrate activation determining which DNA elements are active per cell goal, governed as substrate content under Paper 1’s commitments).

Relationship type: Enabling pair. The two commitments are mutually dependent in multi-level deployments: B1.07 enables B1.01 to operate across structural levels rather than only within each level; B1.01 motivates the specific governance requirement B1.07’s design must satisfy, making the design derivable from the governance principle rather than from functional adequacy alone.

Position in Phase B4: This is the third note in the B4 composition-pair series, following B4.01 and B4.02. The pair formalizes the architectural interdependence between the architecture’s foundational separation commitment and the mechanism that implements that separation across the three-level structure. B4.04 will cover the next pair: B1.02 (three-level structure with relational role membership) and B1.17 (verification substrates as governance over instinct integration).

Adjacent operational notes: B2.30 (harness substrate as expression governor) and B2.31 (DNA activation patterns under expression) formalize the expression mechanism’s implementation details. B2.81 (vertical evolution downward) establishes expression as the downward vertical evolution pathway. B3.08 (Expression Mechanism Bypass) formalizes the anti-pattern when this pair is incomplete.

2. Enabling direction — B1.07 enables B1.01’s cross-level operation

B1.01 commits to separating fast-pattern instinct from deliberate reasoning into independently-evolving layers under unified human governance. The instinct layer is the LLM, operating as the pattern-matched, fast-path component of the cell. The reasoning layer is the CKS substrate inherited from Paper 1 — the persistent, human-governed orchestration and behavior substrate that specifies how the cell functions. At cell scope, the separation is architecturally complete: cell DNA (the reasoning layer) governs what the cell’s LLM (the instinct layer) does. Cell DNA specifies orchestration rules, behavior substrates, conflict-handling logic, and lifecycle policies; the LLM executes pattern-matched responses within the space those rules define.

The three-level structure of Paper 2 — cells composing into aspects, aspects into Selves — creates a governance challenge this cell-scope separation does not by itself address. Self-level reasoning governance (Self DNA) must be able to affect cell-level instinct behavior. Without a governed pathway for that effect, Self governance can only reach cells through direct cell DNA modification: overwriting cell DNA elements, adding new elements, or reconfiguring existing ones outside the expression channel. That path is technically possible but loses the governed channel the expression mechanism provides — the pathway through which Self-scope DNA configuration shapes cell DNA activation

without replacing or overriding cell-level reasoning governance.

B1.07's expression mechanism is precisely that governed pathway. Each cell carries a harness substrate specifying which DNA elements are active for current activity. That harness substrate is itself substrate content — human-governed, fully inspectable, modifiable, and overridable under Paper 1's commitments. Self DNA (the reasoning layer at Self scope) includes harness substrate configuration specifying which DNA elements cells should activate or deactivate under Self-scope governance. When a cell operates, it consults its harness substrate per B1.01 — reasoning governs instinct — and the content of that harness substrate reflects Self-level expression per B1.07. The cell's LLM (instinct) operates under DNA rules shaped by both cell-scope reasoning governance (what cell DNA specifies) and Self-scope reasoning governance propagated through the governed expression channel.

The enabling contribution is a scope extension. B1.01 at cell scope is architecturally self-sufficient for single-level deployments. B1.07 enables B1.01 to remain architecturally coherent at the multi-level scope Paper 2's three-level structure requires. Without B1.07, each cell's instinct/reasoning separation is isolated at cell scope, with no governed pathway for Self-level reasoning governance to extend its reach downward to cell-level instinct behavior. The expression mechanism is what makes the instinct/reasoning separation coherent across structural levels rather than only within each level considered in isolation.

3. Motivating direction — B1.01 motivates B1.07's governance requirement

The expression mechanism exists in the source paper as a mechanism for selecting which DNA-layer substrates activate per goal. That functional role could in principle be satisfied by a purely technical configuration feature: a flag registry, a feature-toggle system, or any other activation-selection mechanism that does not carry the full governance semantics Paper 1 requires. The architectural commitment Paper 2 defends is not that activation selection exists but that activation selection is a governed pathway — substrate content under Paper 1's governance commitments, human-inspectable, human-modifiable, human-overridable, authored under orchestration rules that humans control.

B1.01 is the reason the governance requirement is non-negotiable. If the reasoning layer must govern the instinct layer (B1.01), and if the reasoning layer operates at multiple structural levels (Self, aspect, cell), then any pathway through which higher-level reasoning governance affects lower-level instinct behavior must itself be a governed pathway. An expression pathway that operates outside the governed substrate would mean that Self-scope effects on cell instinct behavior could occur through channels the reasoning layer does not govern — which contradicts B1.01's foundational commitment. The expression mechanism is the cross-level implementation of the instinct/reasoning separation; it must therefore carry the same governance semantics that the separation itself requires.

The design consequence is visible in the expression mechanism's specific architecture. The mechanism activates and deactivates DNA elements — shaping the content of the reasoning layer at cell scope — rather than directly modifying cell LLM behavior. This design choice preserves the cell-level reasoning

governance B1.01 requires: cell DNA still governs cell LLM, the cell's own orchestration rules remain intact, and the expression mechanism operates by determining which orchestration rules are active rather than by replacing or bypassing them. If the expression mechanism instead directly modified cell LLM behavior — bypassing cell DNA — the cell-level instinct/reasoning separation would be compromised: cell instinct behavior would be shaped by something other than the cell's own reasoning layer. B1.01 does not permit that.

B1.01 also explains the expression mechanism's scope in the governance chain. Self DNA governs expression configuration; that configuration shapes cell DNA activation; cell DNA governs cell LLM behavior. The three-step governance chain — Self reasoning → cell reasoning → cell instinct — is B1.01's principle applied recursively across levels. Each step maintains reasoning-layer governance over instinct behavior; no step allows instinct behavior to be shaped outside the reasoning layer at any structural level. The expression mechanism's design and scope are both derivable from B1.01 as the motivating governance principle. Without B1.01, the mechanism could be implemented as any technically adequate activation-selection mechanism; B1.01 is what makes the DNA-element-activation design architecturally mandatory rather than contingently chosen.

4. Mutual dependency

B1.01 depends on B1.07 for cross-level operation. At cell scope, B1.01 is self-sufficient: cell DNA governs cell LLM, and the instinct/reasoning separation holds without assistance from B1.07. In a multi-level deployment, however, Self-level reasoning governance needs to propagate to cell instinct behavior. Without B1.07, the only available path is direct cell DNA modification — which bypasses the governed expression channel and degrades the governance chain from Self scope to cell scope. The cell's instinct behavior then reflects Self-scope governance only through direct DNA writes, outside the governed pathway, with a corresponding loss of the inspectability, traceability, and overridability that governed substrate content provides. B1.07 is the structure that makes B1.01's governance chain coherent across levels rather than only within each level.

B1.07 depends on B1.01 for design rationale. The expression mechanism's specific design — activating and deactivating DNA elements at cell scope rather than modifying LLM behavior directly — is not derivable from the mechanism's functional description alone. Feature-toggle systems, capability flags, and configuration registries all satisfy the functional requirement of determining which content is active for a given goal, without adopting the DNA-element-activation design. The design is derivable from B1.01: because cell-level instinct/reasoning separation must be preserved, the expression mechanism must operate on the reasoning layer's content (DNA elements) rather than on instinct behavior directly. Without B1.01 as the governing principle, the expression mechanism's design is a contingent implementation choice; B1.01 is what makes it an architectural requirement.

The dependency is asymmetric in scope. B1.01 needs B1.07 only in multi-level deployments; single-cell deployments can instantiate B1.01 without B1.07 playing a cross-level role. B1.07's design rationale depends on B1.01 in all deployments, including single-cell ones where expression is

cell-internal. The pair's enabling relationship is strongest and most architecturally consequential in multi-level deployments — which are the deployments Paper 2's three-level structure is principally concerned with.

5. Co-presence requirement

B1.01 without B1.07 — cell-scope-isolated separation. The instinct/reasoning separation operates correctly at each cell: cell DNA governs cell LLM, and each cell's reasoning layer governs its instinct layer. Self-level reasoning governance exists and is architecturally present, but it has no governed downward pathway to cells. Self governance reaches cells only through direct cell DNA modification — writing cell DNA content outside the governed expression channel. The cross-level governance chain from Self reasoning to cell instinct behavior is broken at the pathway level: what passes for cross-level governance is direct DNA manipulation that bypasses the governed channel. The deployment has multiple instances of B1.01 operating at isolated cell scopes but lacks the mechanism that composes them into a coherent multi-level governance chain. The instinct/reasoning separation is locally correct and globally incomplete.

B1.07 without B1.01 — expression without separation principle. The harness substrate configuration mechanism exists and is technically functional: cells activate and deactivate DNA elements based on harness substrate configuration, and Self DNA includes configuration that shapes those activations. But the governance rationale for the mechanism's design is absent. Expression is treated as a deployment-convenience feature — a way to configure which DNA content cells use under current conditions — rather than as the governed channel through which the reasoning layer extends its governance reach downward. Without B1.01 as the motivating governance principle, the expression mechanism may be misconfigured in practice: its harness substrates may not carry the full governance semantics Paper 1 requires; its configuration may not be formally part of Self DNA that reasoning governance authors; its outputs may shape cell behavior without the complete governance chain B1.01 requires at each level. The channel exists technically but has lost the architectural meaning that makes it a governed extension of the reasoning layer rather than a configuration utility.

Full co-presence — governed cross-level reasoning chain. Self DNA (reasoning at Self scope) includes harness substrate configuration per B2.30 specifying which cell DNA elements are active under Self-scope governance. When cells operate, they consult their harness substrates in accordance with B1.01 — reasoning governs instinct. The content of those harness substrates reflects Self-scope reasoning governance propagated through the expression channel. The cell's LLM operates under DNA rules shaped by both cell-scope reasoning governance (its own DNA elements) and Self-scope reasoning governance propagated through expression — with neither level's reasoning governance bypassed. The three-step governance chain (Self reasoning → cell reasoning → cell instinct) holds intact throughout. B1.01's instinct/reasoning separation extends coherently from cell scope to Self scope through the governed channel B1.07 provides, and the separation holds at every structural level because the expression mechanism preserves it at each step of the downward propagation.

6. Operational integration

The operational integration of the two commitments follows a governance chain across substrate layers, beginning at Self scope and terminating at cell instinct behavior.

Self governance per B1.01 at Self scope authors Self DNA per B2.21 and B2.23. Self DNA includes harness substrate configuration per B2.30 specifying which cell DNA elements are activated or deactivated under Self-scope reasoning governance. The harness substrate configuration is substrate content under Paper 1's governance commitments: human-governed, inspectable, modifiable, and overridable. It is not a technical artifact outside the governance framework; it is a first-class substrate element that the reasoning layer at Self scope governs and authors.

Cells operate under their own DNA per B1.01: cell DNA (the reasoning layer at cell scope) governs cell LLM behavior (the instinct layer at cell scope). The set of DNA elements available to activate is the cell's own reasoning layer — its orchestration rules, behavior substrates, conflict-handling logic, and lifecycle policies. What the harness substrate determines is which subset of those elements is active for current activity. The expression mechanism per B1.07 is operating at the moment the cell's harness substrate consults Self-scope configuration to determine which cell DNA elements are active. The cell's LLM then operates within the space those active DNA elements define, which is the instinct/reasoning separation holding at cell scope with cell DNA content shaped by Self-scope expression.

The result is a two-level reasoning governance chain operating through the composition of B1.01 and B1.07. Self-scope reasoning governance (Self DNA) shapes the selection over cell-scope reasoning governance (which cell DNA elements are active). Cell-scope reasoning governance (active cell DNA elements) shapes cell instinct behavior (cell LLM operation). Self-level reasoning governance propagates to cell-level instinct behavior through the governed expression channel, without bypassing either level's reasoning-layer governance and without requiring the direct DNA modification that would characterize Expression Mechanism Bypass (B3.08).

Vertical evolution downward per B2.81 operates through this same composition. When Self governance authors changes to harness substrate configuration — updating which DNA elements cells activate under Self-scope governance — those changes propagate to cell instinct behavior through the governed expression pathway. The expression mechanism is both the cross-level governance channel for B1.01 and the downward pathway for vertical evolution; the two functions are served by the same mechanism because that mechanism's design is shaped by B1.01's governance requirement throughout. Changes to Self DNA that affect expression configuration are traceable through the substrate governance chain, satisfying Paper 1's path retraceability commitment at the cross-level scope.

7. Detection

B2.34 expression mechanism inheritance verification. Is the harness substrate correctly configured as governed substrate content? Does it carry Self-scope governance as its input — authored as part of Self DNA under reasoning-layer governance — or has it been locally configured at cell scope only, outside the governed cross-level channel? Verification confirms that harness substrate configuration is authored under Self-scope reasoning governance and is not a cell-local technical configuration that bypasses that governance.

B2.03 architectural test at Self scope. Does Self-level reasoning governance reach cells through the governed expression pathway? The test passes when the governance chain — Self DNA → harness substrate configuration → cell DNA element activation → cell LLM behavior — is intact and traceable at every step. The test fails when any link in the chain is missing: when Self DNA does not include harness substrate configuration, when harness substrate configuration is not governed substrate content, when cell DNA element activation does not reflect Self-scope expression, or when the LLM operates outside the active DNA elements' scope.

Pair completeness check. Three conditions must hold simultaneously for the pair to be present in full. First: Self DNA includes harness substrate configuration per B2.30. Second: cells' active DNA elements reflect Self-level expression activation — the configuration in Self DNA is operationally shaping which DNA elements cells use. Third: those active DNA elements are themselves governed substrate content traceable to the reasoning-layer authorship B1.01 requires at each structural level. A deployment that satisfies the first condition but not the second has the governed channel authored but not operationally connected; a deployment that satisfies the second but not the third has the channel connected but the cells operating over ungoverned DNA content. All three conditions are required for the composition pair to be architecturally complete.

8. Failure mode

The primary failure mode is Expression Mechanism Bypass (B3.08): Self-level governance reaches cells through direct cell DNA modification rather than through the governed expression pathway. The expression mechanism may exist technically — harness substrates are present, DNA elements can be activated and deactivated — but it is not used for cross-level governance. Instead, Self governance writes cell DNA directly, overwriting or adding elements outside the governed channel. The expression mechanism becomes a cell-internal tool for deployment convenience while cross-level governance operates through direct DNA modification.

This failure mode produces two compounding degradations. The first is the loss of the governed expression channel's governance semantics: direct DNA modification lacks the inspectability, traceability, and overridability that governed harness substrate configuration provides, because the direct modification pathway operates outside the substrate governance framework rather than through it. The second is weakened B1.01 at cross-level scope: the instinct/reasoning separation holds at cell scope (cell DNA still governs cell LLM) but the separation's extension to Self scope is compromised, because Self-scope reasoning governance is reaching cell instinct through a channel — direct DNA

modification — that bypasses the governed expression pathway and its governance semantics.

The failure is detectable through the pair completeness check in §7. If Self DNA does not include harness substrate configuration that operationally shapes cell DNA element activation, the governed expression channel is not present in the cross-level governance chain, and B3.08-class degradation is the operational state. The detection is structural rather than behavioral: the failure is visible in the absence of the governed pathway, not only in observable misbehavior.

A secondary failure mode — expression without B1.01 governance rationale — is the B1.07-without-B1.01 case described in §5. It produces misconfigured expression rather than bypassed expression: the mechanism is used but treated as a configuration convenience rather than as a governed extension of the reasoning layer, with corresponding loss of the governance chain's coherence at each step of propagation.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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